

Association of Adenovirus 36 Infection with Obesity and Metabolic Markers in Humans: A Meta-Analysis of Observational Studies

Background Several studies have shown that Adenovirus 36 (Ad36) influences the risk of obesity in humans. Clarifying the relationship between Ad36 infection and obesity could lead to more effective approaches for the management of obesity. The objective of this study was to conduct a meta-analysis to confirm the influence of Ad36 infection on obesity and metabolic markers.

Methodology/Principal Findings We searched MEDLINE and the Cochrane Library for pertinent articles (including their references) published between 1951 and April 22, 2012. Only English language reports of original observational studies were included in this meta-analysis. Data extraction was performed independently by two reviewers. Weighted mean differences (WMDs) and pooled odds ratios (ORs) with 95% confidence intervals (95% CIs) were calculated using the random effects model. Of 237 potentially relevant studies, 10 cross-sectional studies ($n = 2,870$) conformed to the selection criteria. Pooled analysis showed that the WMD for BMI of Ad36 infection compared with non-infection was 3.19 (95% CI 1.44–4.93; $P < 0.001$). Sensitivity analysis restricted to studies of adults yielded a similar result of 3.18 (95% CI 0.78–5.57; $P = 0.009$). The increased risk of obesity associated with Ad36 infection was also significant (OR: 1.9; 95% CI: 1.01–3.56; $P = 0.047$). No significant differences were found in relation to total cholesterol ($P = 0.83$), triglycerides ($P = 0.64$), HDL ($P = 0.69$), blood glucose ($P = 0.08$), waist circumference ($P = 0.09$), and systolic blood pressure ($P = 0.25$).

Conclusion/Significance Ad36 infection was associated with the risk of obesity and weight gain, but was not associated with abnormal metabolic markers including waist circumference. It suggests that Ad36 infection is more associated with accumulation of subcutaneous fat than that of visceral fat. The relationship between Ad36 and obesity should be assessed by further studies, including well-designed prospective studies, to gain a better understanding of whether Ad36 plays a role in the etiology of human obesity.